FORM HS-06 · REV. 4 POST-STASIS REFRESHER · MODULE 6 OF 9 POST IN EVERY CRANE CAB, EQUIPMENT YARD AND LIFT MUSTER

DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Crane & Lift / Heavy Equipment Pool

NEVER UNDER THE LOAD.

CRANES, RIGGING, HEAVY EQUIPMENT AND TRENCHES. CRUSH AND STRUCK-BY DEATHS HAPPEN IN A SECOND.

RULE ZERO

NEVER STAND, WALK OR WORK UNDER A SUSPENDED LOAD.

Slings snap, hooks fail, and loads swing in gusts. If a load is in the air, you are outside its fall zone. If the load has to pass over you, you leave. Operators: **never swing a load over people.** Stop and wait for the area to clear.

20 ft	30 mph	75%
POWER LINE CLEARANCE (≤350 KV)	ALL LOADS DOWN (SITE RULE)	OF CHART = CRITICAL LIFT

EVERY LIFT, IN ORDER

CRANE & LIFT RUNS ON CH 11

1. **Plan the lift.** Load weight, rigging weight, radius, boom length, crane capacity at that radius. Is it a critical lift? Then it needs an engineered plan signed by the Lift Director (see below).
2. **Inspect.** The operator does the daily crane inspection. Riggers inspect every sling, shackle and hook before use. Damaged gear gets tagged out and removed.
3. **Set up on solid ground.** Loose sand won't hold an outrigger. Full outrigger extension on **crane mats**. Check for buried conduit, voids and fresh backfill.
4. **Barricade the swing radius.** Hard barricade around the rotating counterweight so no one gets crushed between it and anything else.
5. **Assign one signal person** in an orange hard hat, in view of the operator or on Ch 11. Brief the crew: load path, landing zone, who signals.
6. **Test lift.** Lift a few inches, hold, check brakes, balance and rigging. Set it down and fix anything that moves.
7. **Lift with taglines.** Control the load from outside the fall zone with taglines. Never hold a load with your hands, and never ride the hook.

CRITICAL LIFTS

Site definition. Any of these makes a lift critical, and it needs an engineered lift plan, a pre-lift meeting and the Lift Director on site:

- Load is **more than 75% of the crane's chart capacity** at that radius
- Two or more cranes on one load
- Lifting personnel in a platform
- Load passes over occupied structures, habs or live systems
- Unusual loads: high sail area panels, unknown weight, odd center of gravity

POWER LINES AND SITE GRIDS

Minimum distance between any part of the crane, load or rigging and an energized line. Site power grids, bus bars and overhead conduits count.

VOLTAGE	MINIMUM CLEARANCE
Up to 350 kV	20 ft
350 – 1,000 kV	50 ft
Over 1,000 kV	Set by Electrical & Power + engineer

Closer than that? The line gets de-energized and locked out by Electrical & Power, or you don't lift. Use a dedicated spotter whenever you work near the limit.

WIND LIMITS

CH 16 BROADCASTS GUSTS · READ THE CAB ANEMOMETER

0-19	**NORMAL** mph sustained or gusts	Lift per the manufacturer's chart. Many cranes are limited at about 20 mph. **The crane's own limit always wins.**
20-29	**RESTRICTED** Site rule	**No personnel lifts.** High-sail loads (wall panels, hull plate, roof sections) only with Lift Director approval. Extra taglines.
30+	**ALL LOADS DOWN** Site rule	Land every suspended load at 30 mph sustained **or** gusts. Secure the crane per the manufacturer. Sandstorm siren (continuous tone) = booms down, crews to shelter.

RIGGING: REJECT IT IF...

- **No tag or unreadable capacity tag.** No tag, no lift.
- **Wire rope:** broken wires (running rope: 6 random broken wires in one lay, or 3 in one strand in one lay), kinks, crushing, birdcaging, core sticking out, heavy corrosion, heat or weld-spatter damage.
- **Synthetic slings:** cuts, burns, melted or charred spots, exposed red core yarns, UV-faded and stiff. This sun eats synthetics. Store them out of the light.
- **Hooks:** missing or broken safety latch, throat opened up, twisted or cracked.
- **Shackles:** wrong pin, bent body, pin not fully seated.

SLING ANGLE EATS CAPACITY

The flatter the sling, the harder each leg pulls. Multiply each leg's share of the load by the factor.

ANGLE FROM HORIZONTAL	LOAD FACTOR PER LEG
90° (straight up)	1.00 ×
60°	1.15 ×
45°	1.41 ×
30°	2.00 ×

Example: a 10,000 lb panel on two legs at 30°. Each leg carries 5,000 × 2 = **10,000 lb**. Don't rig flatter than 30°.

DO

- Know who the signal person is before the load moves.
- Give **STOP** the moment you see danger. Anyone can give it. The operator obeys it every time.
- Stay outside barricades and the counterweight swing.
- Use taglines and stand clear of pinch points when landing.
- Learn the hand signals.

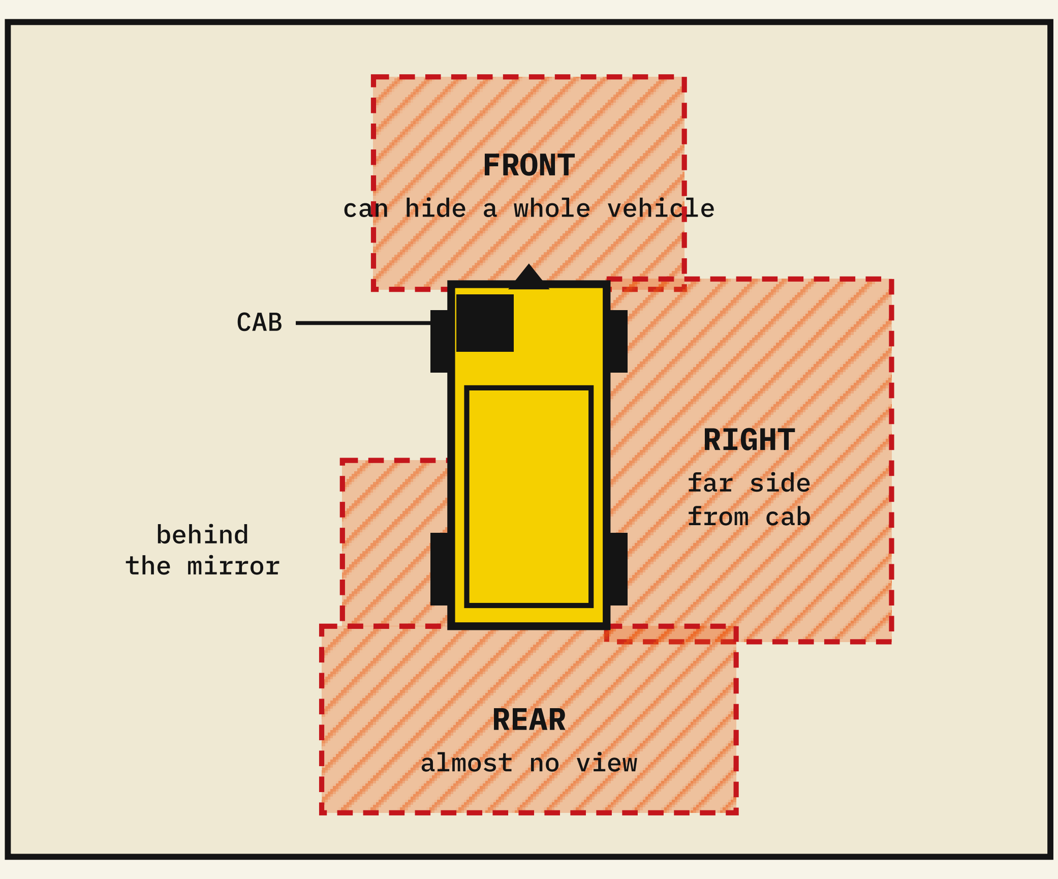
HAND SIGNAL CHART →

DON'T

- Walk under a load, ever.
- Ride a hook, load or headache ball.
- Take signals from more than one person (except STOP).
- Side-load or drag a load with the crane.
- Climb on a crane that's running or about to swing.

HEAVY EQUIPMENT

HEAVY EQUIPMENT POOL RUNS ON CH 12



Haul truck, top view, cab on the left. Shaded areas are where the operator can't see you from the seat. Bigger machines have bigger blind spots.

IF YOU CAN'T SEE THE OPERATOR'S EYES, THEY CAN'T SEE YOU.

Make eye contact and get a wave before you walk near any machine. Approach from the cab side, in their view.

- **Walk-around inspection** before every shift: tires, tracks, leaks, lights, alarm, mirrors, cameras, fire extinguisher.
- **Three points of contact** getting on and off. Face the machine. Never jump down.
- **Seat belt on, every time.** The rollover cage (ROPS) only protects you if you stay inside it.
- **Backup alarm working**, and a spotter whenever you back up somewhere you can't see.
- **Loose sand gives way.** Stay off soft edges, dune faces and fresh fill. Drive straight up and down slopes, never across.
- **Lower the bucket or blade**, set the brake and shut down before you leave the seat.

HORN SIGNALS

1	2	3
STOPPING	MOVING FORWARD	BACKING UP

Operators sound the horn before moving. Hear 2 or 3 blasts and you're near the machine? Get clear and make eye contact.

STRUCK-BY ZONES

- **Hi-vis vest on** outside any cab or shelter, day and night. Sand haze kills visibility.
- Haul roads and loading zones are **exclusion zones**. Walk only on marked crossings.
- Never stand between a machine and a fixed object, or between two machines.
- Stay out of the swing of excavators. If it rotates, assume it will hit you.
- Spotters stand where the operator can see them, never behind the machine.

TRENCHES & EXCAVATIONS

SAND CAVES IN WITHOUT WARNING

OUR SOIL IS TYPE C

Dry, loose sand is the least stable soil class. A cubic yard of it weighs about as much as a car. Walls collapse with no cracks or warning.

5 ft	34°	2 ft
PROTECTIVE SYSTEM REQUIRED	MAX SLOPE (1½ : 1)	SPOILS BACK FROM EDGE

Protective system means **sloping** the walls back to 34° or flatter, **shoring**, or a **trench shield** (box). Deeper than 20 ft needs an engineer's design.

TRENCH RULES

- A **competent person** inspects every excavation before each shift, after every sandstorm or rain, and whenever conditions change.
- **Ladder, ramp or stair within 25 ft** of every worker in trenches 4 ft deep or more.
- Keep spoil piles, tools and equipment **at least 2 ft back** from the edge.
- Never work outside the shield or in an unprotected section, even "just for a minute."
- Keep heavy equipment back from the edge. Vibration and weight start collapses.
- Locate and mark buried power, water and data lines before you dig.
- Test the air in trenches deeper than 4 ft where gas can collect.

CONTACT OR COLLAPSE

A crane or machine touches a power line, a load drops, or a trench caves in.

ALL STOP FIRST. THEN CALL IT IN.

Most trench rescue victims are the would-be rescuers. A second collapse often follows the first.

- **Call "ALL STOP, ALL STOP"** on your channel, then **Ch 1: "Emergency, emergency, emergency"** with the type (Electrical, Struck-by, Caught-in), sector-grid, deck and number hurt.
- **Power line contact:** operator stays in the cab unless it's on fire. If you must get out, jump clear with feet together, never touching machine and ground at once, then shuffle away in small steps. Everyone else stays back at least 35 ft.
- **Trench collapse:** don't jump in to dig. Keep everyone out, shut down nearby equipment, and wait for Fire & Rescue with shoring.
- **Dropped load:** don't lift it off a person with a crane unless Fire & Rescue directs it. Keep the area clear.

ANYONE CAN CALL STOP

Arms out, palms down, wave them back and forth, or say "ALL STOP" on the radio. Every operator stops, every time, whoever gives it. Sort out why after the load is safe.

WORKER ACKNOWLEDGEMENT

I have read this poster. I will never go under a suspended load, I know who signals my lifts, and I know I can stop any lift or machine that puts someone in danger.

NAME	COLONIST ID	CREW / SECTOR	DATE

Adapted from Old Earth standards: OSHA 1926 Subpart CC (Cranes & Derricks in Construction, incl. 1926.1408 power line clearance), OSHA 1926 Subpart P (Excavations), ASME B30.5 (Mobile Cranes) and B30.9 (Slings), OSHA 1926.602 (Earthmoving Equipment).

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